

A Predictive Analytics Framework for Proactive Learning Support and Early Student Intervention

A Hybrid Machine Learning and Knowledge-Based Reasoning Project

Collaborative Final Project

CSST101 – Machine Learning

CSST102 – Knowledge Representation and Reasoning

Submitted by:

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PROJECT OVERVIEW

This project proposal addresses SDG 4 by aiming to ensure inclusive and equitable quality education and promote lifelong learning opportunities for all. Student disengagement and a lack of personalized support (addressing "lack of critical thinking" and "learning enthusiasm") are major barriers. We aim to provide educators with a proactive tool to identify at-risk students and deliver targeted, rule-based interventions, fostering a more inclusive and supportive learning environment.

OBJECTIVES

General Objective:

Student disengagement and high dropout rates remain significant challenges in educational institutions, apparently worsening by large class sizes or remote learning environments. Teachers or Professors frequently lack the scalable, real-time tools to identify students who are falling behind due to difficulty, lack of enthusiasm, or other factors before their performance suffers significantly. This project aims to support both teachers, professors and students by using machine learning to proactively predict at-risk students and using a knowledge-representation system to recommend actionable, personalized learning interventions.

Specific Objectives:

- Analyze the Inputs and Generate a Risk Assessment
- Illustrating a Dashboard for Visualization
- Provide Data-Driven recommendations for Support Actions

SYSTEM ARCHITECTURE

User Input → Machine Learning Model → KRR Rules → Final Risk Level → Recommendations

MACHINE LEARNING COMPONENT (CSST101)

Algorithm Used: Random Forest

Classifier

Dataset Size: 201 records (after data cleaning)

Accuracy: 1.0

MACHINE LEARNING PIPELINE

Data Collection:

Student academic, engagement, and behavioral data collected from institutional records and surveys.

Data Preprocessing:

- Removal of incomplete or inconsistent records
 - Encoding of categorical variables
 - Feature selection based on relevance
 - Data cleaning and normalization
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Model Training:

Model Evaluation:

Random Forest model trained using cleaned and encoded student data in Google Colab.

Model Deployment:

Evaluation performed using accuracy score, classification report, and confusion matrix.

DATASET DESCRIPTION

Dataset Type: Structured Tabular
Dataset

Number of Records: 201

Target Variable: Student Risk Level
(Low, Moderate, High)

KNOWLEDGE REPRESENTATION & REASONING (CSST102)

Rule 1:

IF Attendance < 60 AND Engagement_Level is Low
THEN Risk Level = High

Rule 2:

IF Attendance > 90 AND Engagement_Level is High
THEN Risk Level = Low

Rule 3:

IF Quiz_Performance_Average $\geq$ 85 AND Attendance $\geq$ 90
THEN Risk Level = Low

Rule 4:

IF Attendance < 70 AND Engagement_Level is Moderate
THEN Risk Level = Moderate

Rule 5:

IF none of the rules apply
THEN use Machine Learning Prediction

HYBRID DECISION LOGIC

The system combines probabilistic predictions from the machine learning model with deterministic rule-based logic.

The ML model provides an initial prediction, which is then adjusted using KRR rules to improve interpretability, handle edge cases, and ensure logical consistency in decision-making.

SYSTEM FEATURES

- ✓ Wellness risk prediction
- ✓ Rule-based recommendations
- ✓ Web interface / API (Flask)
- ✓ Google Colab deployment

TESTING AND EVALUATION

Test Case	Input Summary	Expected Output
1	Low attendance, low engagement	High Risk
2	High attendance, high engagement	Low Risk
3	Average attendance and engagement	Moderate Risk
4	User input via web form	Real-time risk prediction

CONCLUSION

This project successfully demonstrates a hybrid predictive analytics system that integrates machine learning with knowledge representation and reasoning. By combining a Random Forest classifier with rule-based logic, the system achieves high predictive accuracy while maintaining interpretability. The deployment of the model through a Flask web application allows real-time user interaction and practical application in academic risk assessment.

GROUP CONTRIBUTION

Carl Henrik M. Cipriano | Programmer, Model Testing/Analysis & Documentation
Janvher Lucas J. Sarmiento | Programmer & Web Deployment
